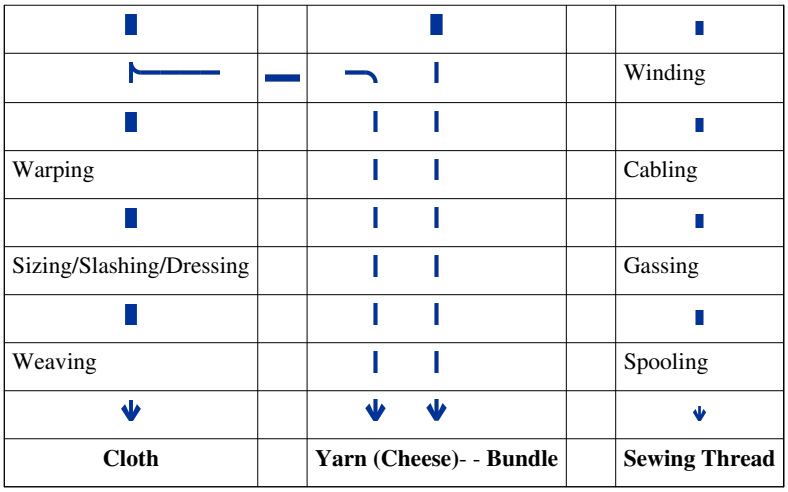
Textile manufacturing

Textile manufacturing is a major industry. It is based in the conversion of three types of fibre into yarn, then fabric, then textiles. These are then fabricated into clothes or other artifacts. Cotton remains the most important natural fibre, so is treated in depth. There are many variable processes available at the spinning and fabric-forming stages coupled with the complexities of the finishing and colouration processes to the production of a wide ranges of products. There remains a large industry that uses hand techniques to achieve the same results.

Processing of cotton

Cotton Manufacturing Processes (after Murray 1911)

Bale Breaker				**Blowing Room**
Willowing				
Breaker Scutcher		Batting		
Finishing Scutcher		Lapping		
Carding				**Carding Room**
Silver Lap				
Combing				
Drawing				
Slubbing				
Intermediate				
Roving		Fine Roving		
Mule Spinning	-	Ring Spinning		**Spinning**
		Reeling		Doubling
Winding		Bundling		Bleaching



Cotton is the world's most important natural fibre. In the year 2007, the global yield was 25 million tons from 35 million hectares cultivated in more than 50 countries.[1]

There are five stages[2]

- Cultivating and Harvesting
- Preparatory Processes
- Spinning
- Weaving
- Finishing

Cultivating and harvesting

Cotton is grown anywhere with long, hot dry summers with plenty of sunshine and low humidity. Indian cotton, gossypium arboreum, is finer but the staple is only suitable for hand processing. American cotton, gossypium hirsutum, produces the longer staple needed for machine production.[3] Planting is from September to mid November and the crop is harvested between March and May. The cotton bolls are harvested by stripper harvesters and spindle pickers, that remove the entire boll from the plant. The cotton boll is the seed pod of the cotton plant, attached to each of the thousands of seeds are fibres about 2.5 cm long.[4]

- **Ginning**

  The seed cotton goes in to a Cotton gin. The cotton gin separates seeds and removes the "trash" (dirt, stems and leaves) from the fibre. In a saw gin, circular saws grab the fibre and pull it through a grating that is too narrow for the seeds to pass. A roller gin is used with longer staple cotton. Here a leather roller captures the cotton. A knife blade, set close to the roller, detaches the seeds by drawing them through teeth in circular saws and revolving brushes which clean them away.[5]

  The ginned cotton fibre, known as lint, is then compressed into bales which are about 1.5 m tall and weigh almost 220 kg. Only 33% of the crop is usable lint. Commercial cotton is priced by quality, and that broadly relates to the average length of the staple, and the variety of the plant. Longer staple cotton (2½ in to 1¼ in) is called Egyptian, medium staple (1¼ in to ¾ in) is called American upland and short staple (less than ¾ in) is called Indian.[6]

  The cotton seed is pressed into a cooking oil. The husks and meal are processed into animal feed, and the stems into paper.

Issues

Cotton is farmed intensively and uses large amounts of fertiliser and 25% of the worlds insecticide. Native Indian variety were rainwater fed, but modern hybrids used for the mills need irrigation, which spreads pests. The 5% of cotton-bearing land in India uses 55% of all pesticides used in India.[3] Before mechanisation, cotton was harvested manually and this unpleasant task was done by the lower castes, and in the United States by slaves of African origin.

Preparatory processes- preparation of yarn

- **Ginning, bale-making and transportation** is done in the country of origin.
- **Opening and cleaning**



Platt Bros. Picker

Cotton mills get the cotton shipped to them in large, 500 pound bales. When the cotton comes out of a bale, it is all packed together and still contains vegetable matter. The bale is broken open using a machine with large spikes. It is called an **Opener**.In order to fluff up the cotton and remove the vegetable matter, the cotton is sent through a picker, or similar machines. A **picker** looks similar to the carding machine and the cotton gin, but is slightly different. The cotton is fed into the machine and gets beaten with a beater bar, to loosen it up. It is fed through various rollers, which serve to remove the vegetable matter. The cotton, aided by fans, then collects on a screen and gets fed through more rollers till it emerges as a continuous soft fleecy sheet, known as a lap.[6] [7]

- **Blending**,

Mixing & Scutching

Scutching refers to the process of cleaning cotton of its seeds and other impurities. A scutching machine for cotton was first invented in 1797, but didn't get much attention until it was introduced in Manchester in 1808 or 1809. By 1816 it had been generally adopted.[5] The scutching machne worked by passing the cotton through a pair of rollers, and then striking it with iron or steel bars called beaters. The beaters, which turn very quickly, strike the cotton hard and knock the seeds out. This process is done over a series of parallel bars so as to allow the seeds to fall through. At the same time a breeze is blown across the bars, which carries the cotton into a cotton chamber.

- **Carding**

Carding: the fibres are separated and then assembled into a loose strand (sliver or tow) at the conclusion of this stage.



Carding machine

The cotton comes off of the picking machine in laps, and is then taken to carding machines. The carders line up the fibres nicely to make them easier to spin. The carding machine consists mainly of one big roller with smaller ones surrounding it. All of the rollers are covered in small teeth, and as the cotton progresses further on the teeth get finer (i.e. closer together). The cotton leaves the carding machine in the form of a sliver; a large rope of fibres.[8]

Note: In a wider sense Carding can refer to these four processes: Willowing- loosening the fibres; Lapping- removing the dust to create a flat sheet or lap of cotton; Carding- combing the tangled lap into a thick rope of 1/2 in in diameter, a sliver; and Drawing- where a drawing frame combines 4 slivers into one- repeated for increased quality.

- **Combing** is optional,but is used to remove the shorter fibres, creating a stronger yarn.

[9]



A Combing machine

- **Drawing** the fibres are straightened

  Several slivers are combined. Each sliver will have thin and thick spots, and by combining several slivers together a more consistent size can be reached. Since combining several slivers produces a very thick rope of cotton fibres, directly after being combined the slivers are separated into rovings. These rovings (or slubbings) are then what are used in the spinning process.[10]

  Generally speaking, for machine processing, a roving is about the width of a pencil.

  - Drawing frame: Draws the strand out
  - Slubbing Frame: adds twist, and winds on to bobbins
  - Intermediate Frames: are used to repeat the slubbing process to produce a finer yarn.
  - Roving frames: reduces to a finer thread, gives more twist, makes more regular and even in thickness, and winds on to a smaller tube.[11]

Spinning- yarn manufacture

- **Spinning**

  The spinning machines take the roving, thins it and twists it, creating yarn which it winds onto a bobbin.[12]

  In mule spinning the roving is pulled off a bobbin and fed through some rollers, which are feeding at several different speeds.This thins the roving at a consistent rate. If the roving was not a consistent size, then this step could cause a break in the yarn, or could jam the machine. The yarn is twisted through the spinning of the bobbin as the carriage moves out, and is rolled onto a cop as the carriage returns. Mule spinning produces a finer thread than the less skilled ring spinning.[13]

  - The mule was an intermittent process, as the frame advanced and returned a distance of 5ft.It was the descendant of 1779 Crompton device. It produces a softer less twisted thread that was favoured for fines and for weft. It requires considerable skill, so was women's work.
  - The ring was a descendant of the Arkwright water Frame 1769. It was a continuous process, the yard was coarser, had a greater twist and was stronger so was suited to be warp. Requiring less skill it was men's work. Ring spinning is slow due to the distance the thread must pass around the ring, other methods have been introduced. These are collectively known as Break or Open-end spinning.[14]

    Sewing thread, was made of several threads twisted together, or doubled.

- **Checking**

  This is the process where each of the bobbins is rewound to give a tighter bobbin.

- **Folding and twisting**

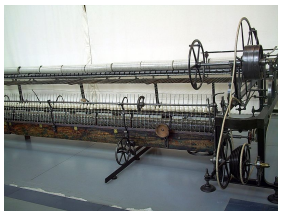
  Plying is done by pulling yarn from two or more bobbins and twisting it together, in the opposite direction that in which it was spun. Depending on the weight desired, the cotton may or may not be plied, and the number of strands twisted together varies.[15]

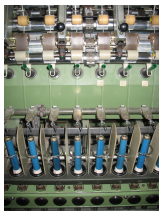
- **Gassing**

  Gassing is the process of passing yarn, as distinct from fabric very rapidly through a series of Bunsen gas flames in a gassing frame, in order to burn off the projecting fibres and make the thread round and smooth and also brighter. Only the better qualities of yarn are gassed, such as that used for voiles, poplins, venetians, gabardines, many Egyptian qualities, etc. There is a loss of weight in gassing, which

varies' about 5 to 8 per cent., so that if a 2/60's yarn is required 2/56's would be used. The gassed yarn is darker in shade afterwards, but should not be scorched.[16]


Mule spinning


Mule spinning


Ring spinning


Ring spinning

Measurements

- Cotton Counts: The number of pieces of thread, 840 yards long needed to make up 1 lb weight. 10 count cotton means that 10x840 yd weighs 1 lb. This is coarser than 40 count cotton where 40x840 yards are needed. In the United Kingdom, Counts to 40s are coarse (Oldham Counts), 40 to 80s are medium counts and above 80 is a fine count. In the United States ones to 20s are coarse counts.
- Hank: A length of 7 leas or 840 yards
- Thread: A length of 54 in (the circumference of a warp beam)
- Bundle: Usually 10 lb
- Lea: A length of 80 threads or 120 yards[17]
- Denier: this is an alternative method. It is defined as a number that is equivalent to the weight in grams of 9000m of a single yarn. 15 denier is finer than 30 denier.
- Tex: is the weight in grams of 1 km of yarn.[18]

The worsted hank is only 560 yd [19]

Weaving-fabric manufacture

The weaving process uses a loom. The lengthway threads are known as the warp, and the cross way threads are known as the weft. The warp which must be strong needs to be presented to loom on a warp beam. The weft passes across the loom in a shuttle, that carries the yarn on a pirn. These pirns are automatically changed by the loom. Thus, the yarn needs to be wrapped onto a beam, and onto pirns before weaving can commence. [20]

- **Winding**

  After being spun and plied, the cotton thread is taken to a warping room where the winding machine takes the required length of yarn and winds it onto warpers bobbins

- **Warping or beaming**

  Racks of bobbins are set up to hold the thread while it is rolled onto the warp bar of a loom. Because the thread is fine, often three of these would be combined to get the desired thread count.[citation needed].


A Warper

- **Sizing**

  Slasher sizing machine needed for strengthening the warp by adding starch to reduce breakage of the yarns .

- **Drawing in, Looming**

  The process of drawing each end of the warp separately through the dents of the reed and the eyes of the healds, in the order indicated by the draft.

- **Pirning (Processing the weft)**

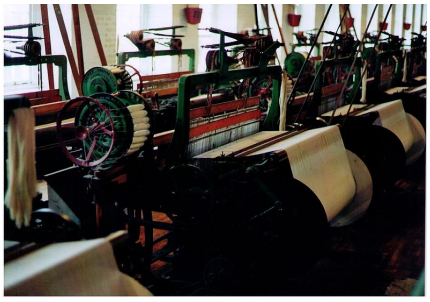
  Pirn winding frame was used to transfer the weft from cheeses of yarn onto the pirns that would fit into the shuttle

- **Weaving**

  At this point, the thread is woven. Depending on the era, one person could manage anywhere from 3 to 100 machines. In the mid nineteenth century, four was the standard number. A skilled weaver in 1925 would run 6 Lancashire Looms. As time progressed new mechanisms were added that stopped the loom any time something went wrong. The mechanisms checked for such things as a broken warp thread, broken weft thread, the shuttle going straight across, and if the shuttle was empty. Forty of these Northrop Looms or automatic looms could be operated by one skilled worker.[21]

  The three primary movements of a loom are shedding, picking, and beating-up.

  - *Shedding*: The operation of dividing the warp into two lines, so that the shuttle can pass between these lines. There are two general kinds of sheds-"open" and "closed." Open Shed-The warp threads are moved when the pattern requires it-from one line to the other. Closed Shed-The warp threads are all placed level in one line after each pick.
  - *Picking*:The operation of projecting the shuttle from side to side of the loom through the division in the warp threads. This is done by the overpick or underpick motions. The overpick is suitable for quick-running looms, whereas the underpick is best for heavy or slow looms.
  - *Beating-up*: The third primary movement of the loom when making cloth, and is the action of the reed as it drives each pick of weft to the fell of the cloth.[22]

  A Draper loom in textile museum, Lowell, Massachusetts

  The Lancashire Loom was the first semi-automatic loom. Jacquard looms and Dobby looms are looms that have sophisticated methods of shedding. They may be separate looms, or mechanisms added to a plain loom. A Northrop Loom was fully automatic and was mass produced between 1909 and the mid 1960s. Modern looms run faster and do not use a shuttle: there are air jet looms, water jet looms and rapier looms.

Measurements

- Ends and Picks: Picks refer to the weft, ends refer to the warp. The coarseness of the cloth can be expressed as the number of picks and ends per quarter inch square, or per inch square. Ends is always written first. For example: *Heavy domestics are made from coarse yarns, such as 10's to 14's warp and weft, and about 48 ends and 52 picks.*[23]

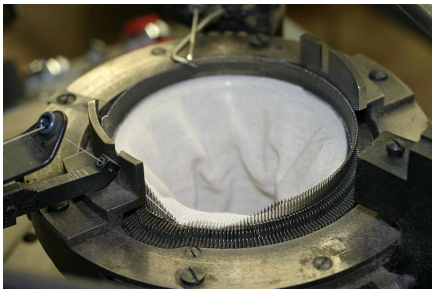
Associated job titles

- Piecer
- Scavenger
- Weaver
- Tackler
- Draw boy
- Pirner

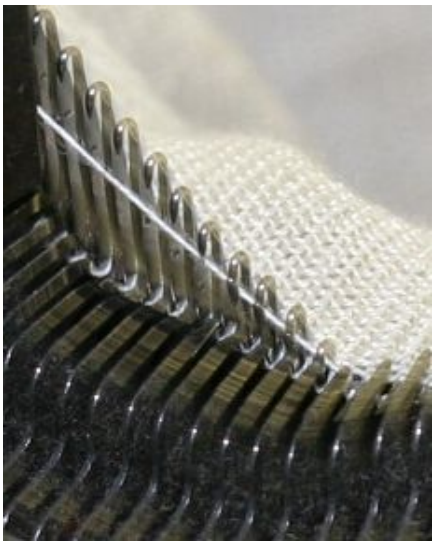
Issues

When a hand loom was located in the home, children helped with the weaving process from an early age. Piecing needs dexterity, and a child can be as productive as an adult. When weaving moves from the home to the mill, children were often allowed to *help* their older sisters, and laws have to be made to prevent child labour becoming established,

Knitting- fabric manufacture



A circular knitting machine.



Close-up on the needles.

Knitting by machine is done in two different ways; warp and weft. Weft knitting (as seen in the pictures) is similar in method to hand knitting with stitches all connected to each other horizontally. Various weft machines can be configured to produce textiles from a single spool of yarn or multiple spools depending on the size of the machine cylinder (where the needles are bedded). In a warp knit there are many pieces of yarn and there are vertical chains, zigzagged together by crossing the yarn.

Warp knits do not stretch as much as a weft knit, and it is run-resistant. A weft knit is not run-resistant, but stretches more. This is especially true if spools of Lycra are processed from separate spool containers and interwoven through the cylinder with cotton yarn, giving the finished product more flexibility and making it less prone to having a 'baggy' appearance. The average t-shirt is a weft knit.[24]

Finishing- processing of textiles

The **grey cloth**,woven cotton fabric in its loom-state, not only contains impurities, including warp size, but requires further treatment in order to develop its full textile potential. Furthermore, it may receive considerable added value by applying one or more finishing processes.[25] [26]

- **Desizing**

  Depending on the size that has been used, the cloth may be steeped in a dilute acid and then rinsed, or enzymes may be used to break down the size.[27]

- **Scouring**

  Scouring, is a chemical washing process carried out on cotton fabric to remove natural wax and non-fibrous impurities (eg the remains of seed fragments) from the fibres and any added soiling or dirt. Scouring is usually carried in iron vessels called kiers. The fabric is boiled in an alkali, which forms a soap with free fatty acids. (saponification). A kier is usually enclosed, so the solution of sodium hydroxide can be boiled under pressure, excluding oxygen which would degrade the cellulose in the fibre. If the appropriate reagents are used, scouring will also remove size from the fabric although desizing often precedes scouring and is considered to be a separate process known as fabric preparation. Preparation and scouring are prerequisites to most of the other finishing processes. At this stage even the most naturally white cotton fibres are yellowish, and bleaching, the next process, is required.[27]

- **Bleaching**

Bleaching improves whiteness by removing natural coloration and remaining trace impurities from the cotton; the degree of bleaching necessary is determined by the required whiteness and absorbency. Cotton being a vegetable fibre will be bleached using an oxidizing agent, such as dilute sodium hydrochlorite or dilute hydrogen peroxide. If the fabric is to be dyed a deep shade, then lower levels of bleaching are acceptable, for example. However, for white bed sheetings and medical applications, the highest levels of whiteness and absorbency are essential.[28]

- **Mercerising**

  A further possibility is mercerizing during which the fabric is treated with caustic soda solution to cause swelling of the fibres. This results in improved lustre, strength and dye affinity. Cotton is mercerized under tension, and all alkali must be washed out before the tension is released or shrinkage will take place. Mercerizing can take place directly on grey cloth, or after bleaching.[29]

  Many other chemical treatments may be applied to cotton fabrics to produce low flammability, crease resist and other special effects but four important non-chemical finishing treatments are:

- **Singeing**

  Singeing is designed to burn off the surface fibres from the fabric to produce smoothness. The fabric passes over brushes to raise the fibres, then passes over a plate heated by gas flames.

- **Raising**

  Another finishing process is raising. During raising, the fabric surface is treated with sharp teeth to lift the surface fibres, thereby imparting hairiness, softness and warmth, as in flannelette.

- **Calendering**

  Calendering is the third important mechanical process, in which the fabric is passed between heated rollers to generate smooth, polished or embossed effects depending on roller surface properties and relative speeds.

- **Shrinking (Sanforizing)**

  Finally, mechanical shrinking (sometimes referred to as sanforizing), whereby the fabric is forced to shrink width and/or lengthwise, creates a fabric in which any residual tendency to shrink after subsequent laundering is minimal.

- **Dyeing**

  Finally, cotton is an absorbent fibre which responds readily to colouration processes. Dyeing, for instance, is commonly carried out with an anionic direct dye by completely immersing the fabric (or yarn) in an aqueous dyebath according to a prescribed procedure. For improved fastness to washing, rubbing and light, other dyes such as vats and reactives are commonly used. These require more complex chemistry during processing and are thus more expensive to apply.

- **Printing**

  Printing, on the other hand, is the application of colour in the form of a paste or ink to the surface of a fabric, in a predetermined pattern. It may be considered as localised dyeing. Printing designs on to already dyed fabric is also possible.

Economic, environmental and political consequences of cotton manufacture

The growth of cotton is divided into two segments i.e. organic and genetically modified.[1] . Cotton crop provides livelihood to millions of people but its production is becoming expensive because of high water consumption, use of expensive pesticides, insecticides and fertiliser. GM products aim to increase disease resistance and reduce the water required. The organic sector was worth $583 million. GM cotton, in 2007, occupied 43% of cotton growing areas..[3]

The consumption of energy in form of water and electricity is relatively high, especially in processes like washing, de-sizing, bleaching, rinsing, dyeing, printing, coating and finishing. Processing is time consuming. The major portion of water in textile industry is used for wet processing of textile (70 per cent). Approximately 25 per cent of energy in the total textile production like fibre production, spinning, twisting, weaving, knitting, clothing manufacturing etc. is used in dyeing. About 34 per cent of energy is consumed in spinning, 23 per cent in weaving, 38 per cent in chemical wet processing and five per cent in miscellaneous processes. Power dominates consumption pattern in spinning and weaving, while thermal energy is the major factor for chemical wet processing.[1] [30]

Processing of other vegetable fibres- other processes

Flax

Flax is a bast fibre, which means it comes in bundles under the bark of the Linum usitatissimum plant. The plant flowers and is harvested.

- Retting
- Breaking
- Scutching
- Hackling or combing

It is now treated like cotton.[31]

Jute

Jute is a bast fibre, which comes from the inner bark of the plants of the Corchorus genus. It is retted like flax, sundried and baled. When spinning a small amount of oil must be added to the fibre. It can be bleached and dyed. It was used for sacks and bags but is now used for the backing for carpets.[32]

Now a recent days Jute-Cotton blended fabrics production processes is innovated by Bangladeshi scientists where Jute fibre are blended with cotton and fabricated for the production of various diversified products such as Home Textiles and different decorative fabrics. Dr.Siddique Ullah, Dr. Monjur-e-Khuda and others from BCSIR and BJRI (Bangladeshi Jute Research Institute) improve the chemical treatment method. In addition high value added shutting shirting are made from this fabric with definite woolenising, bleaching, dyeing, printing and finishing. This Jute-Cotton blended fabrics, in the 1970s, were known as Jutton fabrics. [33]

Hemp

Hemp is a bast fibre from the inner bark of Cannabis sativa. It is difficult to bleach, it is used for making cord and rope.

- Retting
- Separating
- Pounding[34]

Other bast fibres

These bast fibres can also be used: kenaf, urena, ramie, nettle.

Other leaf fibres

Sisal is the main leaf fibre used; others are: abacá and henequen.

Processing of protein fibres

Wool

Wool comes from domesticated sheep. It forms two products, woolens and worsteds. The sheep has two sorts of wool and it in the inner coat that is used. This can be mixed with wool that has been recovered from rags. Shoddy is the term for recovered wool that is not matted, while mungo comes from felted wool. Extract is recovered chemically from mixed cotton/wool fabrics.

The fleece is cut in one piece from the sheep.This is then skirted to remove the soiled wool, and baled. It is graded into long wool where the fibres can be up to 15 in, but anything over 2.5 inches is suitable for combing into worsteds. Fibres less than that form short wool and are described as clothing or carding wool.

At the mill the wool is scoured in a detergent to remove grease (the yolk) and impurities. This is done mechanically in the opening machine. Vegetable matter can be removed chemically using sulfuric acid (carbonising). Washing uses a solution of soap and sodium carbonate. The wool is oiled before carding or combing.

- Woollens: Use noils from the worsted combs, mungo and shoddy and new short wool
- Worsteds

  Combing: Oiled slivers are wound into laps, and placed in the circular comber. The worsted yarn gathers together to form a top. The shorter fibres or noils remain behind and are removed with a knife.
- Angora

Silk

The processes in silk production are similar to those of cotton but take account that reeled silk is a continuous fibre. The terms used are different.

- Opening bales. Assorting skeins: where silk is sorted by colour, size and quality, scouring: where the silk is washed in water of 40 degrees for 12 hours to remove the natural gum, drying: either by steam heating or centrifuge, softening: by rubbing to remove any remaining hard spots.
- Silk throwing (winding). The skeins are placed on a reel in a frame with many others. The silk is wound onto spools or bobbins.
  - Doubling and twisting. The silk is far too fine to be woven, so now it is doubled and twisted to make the warp, known as organzine and the weft, known as tram. In organzine each single is given a few twists per inch (tpi), and combine with several other singles counter twisted hard at 10 to 14 tpi. In tram the two singles are doubled

with each other with a light twist, 3 to 6 tpi. Sewing thread is two tram threads, hard twisted, and machine-twist is made of three hard-twisted tram threads. Tram for the crepe process is twisted at up to 80 tpi to make it 'kick up'.
    - Stretching. The thread is tested for consistent size. Any uneven thickness is stretched out. The resulting thread is reeled into containing 500 yd to 2500 yd. The skeins are about 50 in in loop length.
    - Dyeing: the skeins are scoured again, and discoloration removed with a sulphur process. This weakens the silk. The skeins are now tinted or dyed. They are dried and rewound onto bobbins, spools and skeins. Looming, and the weaving process on power looms is the same as with cotton.
- Weaving. The organzine is now warped. This is a similar process to in cotton. Firstly, thirty threads or so are wound onto a warping reel, and then using the warping reels, the threads are beamed. A thick layer of paper is laid between each layer on the beam to stop entangling.[35]

Processing of man made fibres

Discussion of types of man made fibres

Synthetic fibres are the result of extensive development by scientists to improve upon the naturally occurring animal and plant fibres. In general, synthetic fibers are created by forcing, or extruding, fibre forming materials through holes (called spinnerets) into the air, thus forming a thread. Before synthetic fibres were developed, cellulose fibers were made from natural cellulose, which comes from plants.

The first artificial fibre, known as art silk from 1799 onwards, became known as viscose around 1894, and finally rayon in 1924. A similar product known as cellulose acetate was discovered in 1865. Rayon and acetate are both artificial fibres, but not truly synthetic, being made from wood. Although these artificial fibres were discovered in the mid-nineteenth century, successful modern manufacture began much later in the 1930s. Nylon, the first synthetic fibre, made its debut in the United States as a replacement for silk, and was used for parachutes and other military uses.

The techniques used to process these fibres in yarn are essentially the same as with natural fibres, modifications have to be made as these fibers are of great length, and have no texture such as the scales in cotton and wool that aid meshing.

See also

- Clothing technology
- Glossary of textile manufacturing
- Description of Bangladeshi Textile Engineering course
- Cotton mill
- Timeline of clothing and textiles technology
- Textile manufacture during the Industrial Revolution
- Dref Friction Spinning
- Spinning Wheel
- Spinning
- Open End Spinning
- Carding
- Cotton-Spinning Machinery

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[9] Collier 1970, p. 69

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[12] Collier 1970, pp. 71

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